

Symbiotic heat sharing in MOF-based atmospheric water generators for battery thermal management and sorbent regeneration

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Atmospheric water harvesting (AWH) is limited by the high energy demand for sorbent regeneration, while battery thermal management dissipates substantial low-grade heat unused. Here, we propose a thermally symbiotic architecture integrating these systems. Battery discharge waste heat regenerates a MOF-303 sorbent layer via desorption, while the sorption process passively stabilizes battery temperature. The sorbent's regeneration enthalpy aligns with the battery's thermal output, triggering desorption under -12 to -20 A discharge. Within 46 min, the MOF exceeds 50 °C, releasing vapor for condensation in a sealed setup. Concurrently, sorption extracts heat from the battery, reducing peak temperatures by ~ 10 °C. Notably, this integrated system achieves a water production rate of 1.39 g g⁻¹ day⁻¹, a 117% increase over solar-driven counterparts. Finite-element simulations validate the heat transfer dynamics and identify key thermal efficiency parameters. This work establishes a closed-loop thermal cycle that repurposes energy-storage waste heat to power AWH, enabling sorbent regeneration without solar heating. The strategy offers a pathway to enhance thermal efficiency across various energy and environmental systems through internal heat recovery.

In nature, symbiotic relationships represent finely tuned mutualistic interactions, wherein two distinct organisms co-evolve to establish interdependent systems that improve survival, resilience, and ecological fitness. For example, the human gastrointestinal tract hosts

trillions of microorganisms that facilitate nutrient breakdown, synthesize essential vitamins, and provide a protective barrier against pathogens. In return, these microbes inhabit a nutrient-rich, stable environment supported by the host^{1,2}. A similarly interdependent

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model is exemplified in the interaction between bees and flowering plants: bees harvest nectar and pollen for nourishment, while concurrently mediating pollination essential for plant reproduction^{3,4}. These interactions profoundly influence species persistence by creating energetically favorable, self-sustaining ecosystems^{5,6}. This conceptual foundation of symbiosis, although biological in origin, can be extended to physical and engineering systems that exhibit complementary functional roles.

In engineered thermal systems, symbiotic coupling of components with opposing thermal requirements offers an opportunity to enable efficient energy utilization. Lithium-ion batteries—compared to traditional battery chemistries such as nickel–cadmium, nickel–metal hydride, and lead–acid—provide a superior combination of high energy density, long cycle life, minimal self-discharge, high voltage, low maintenance, and rapid charging^{7,8}. However, lithium batteries generate significant heat during discharge due to internal resistance and electrochemical reactions, which can lead to thermal runaway once it rises to overtemperature^{9–11}. In this hazardous condition, the battery's temperature increases uncontrollably, potentially causing fires or explosions^{12,13}. To mitigate this risk, battery thermal management (BTM) is widely employed to enhance the battery's heat dissipation capabilities, which is crucial for maintaining optimal operating temperatures below 55 °C and ensuring safety, performance, and longevity^{13,14}. Based on the cooling medium, BTM can be classified into air cooling, liquid cooling, and composite phase change material (CPCM) cooling BTMs^{15,16}. While effective in limiting thermal buildup, these cooling strategies are inherently dissipative, discarding the battery's thermal energy into the environment without functional recovery, thus representing a major limitation in thermal resource utilization.

In contrast to BTM systems that require heat extraction, certain emerging environmental technologies depend on continuous heat input to function. One such application is sorption-based AWH, which has gained attention for its ability to extract water vapor from ambient air and convert it into liquid water^{17–19}. This technology captures atmospheric moisture even at low relative humidity (RH) and converts it into potable water, typically using solar irradiance for desorption^{20,21}. A key advantage of sorption-based AWH is its geographical adaptability and scalability. The performance of these systems hinges on the properties of the sorbent materials, primarily MOFs, covalent organic frameworks (COFs)²², zeolites²³, silica gel²⁴, hygroscopic salts²⁵, hydrogels^{26,27}, and their hybrids^{28–31}. Among these, MOFs such as MOF-303³², MOF-801³³, Al-Fum³¹, MIL-160³⁴, and UiO-66³⁵ stand out due to their high surface area, tunability, low regeneration temperature, and exceptional selectivity, enabling adsorption at sub-30% RH conditions^{36,37}. Recent advances in continuous-operation AWH devices have enabled multiple adsorption–desorption cycles per day, dramatically improving water output.^{33,38,39} However, in regions lacking consistent sunlight or where solar irradiance is insufficient, such systems struggle to maintain continuous regeneration, prompting alternative strategies that require externally paid thermal input^{40,41}. This approach may not be viable in resource-limited arid environments, where both water scarcity and energy access are major constraints³⁷.

Interestingly, the thermal demands of BTM and AWH systems are inherently complementary: BTM produces low-grade heat that is typically dissipated, while AWH systems require precisely such low-grade thermal energy for desorption. The operating temperature range of lithium batteries during high-rate discharge⁴² coincides well with the initial regeneration temperatures of water-saturated MOFs used in AWH⁴³, creating a thermodynamic basis for coupling the two. In a conceptual BTM–AWH hybrid system, thermal energy generated during battery discharge can be redirected to regenerate the sorbent through passive heat transfer. In an integrated BTM–AWH system, thermal energy released during battery discharge is harnessed to drive desorption in the sorbent layer, while the sorbent simultaneously acts

as a thermal sink, which absorbs latent and sensible heat during moisture uptake, reducing peak battery temperature and enabling passive regeneration. This bidirectional exchange, in which the thermal output of one subsystem drives the operation of the other while simultaneously receiving stabilizing feedback, establishes a closed-loop, self-regulating partnership. This integrated approach is termed “Thermal symbiosis”. It extends beyond traditional unidirectional waste-heat recovery by forming a mutually reinforcing system that enhances thermal management and functional output without a dedicated external heat source for sorbent regeneration.

Despite this conceptual promise, there have been no prior demonstrations of such a thermally symbiotic strategy, nor established frameworks for integrating AWH and BTM subsystems through direct thermal coupling. To address this gap, we developed a prototype system that couples a lithium battery pack with a passive sorption-based atmospheric water generator (AWG) containing a MOF-303 layer. This architecture enables bidirectional thermal interaction: waste heat from battery discharge is transferred through a thermally conductive copper-based interface to drive MOF regeneration, while the sorbent layer simultaneously buffers temperature spikes during cycling. Experimental results show that at a discharge current of −16 A, the MOF layer passively reaches temperatures above 50 °C within 46 min, initiating effective desorption and moisture release. Concurrently, the sorption module extracts both latent and sensible heat, reducing the battery's peak temperature by approximately 10 °C and allowing safe operation under elevated current loads. To overcome the mismatch in temporal operating profiles between battery discharge and sorbent regeneration, we implement an asynchronous strategy in which three battery modules alternately supply heat to a common AWG unit, enabling up to 15 regeneration cycles per day without relying on solar heating. Notably, the 24 h continuous operation enabled a water production rate of $1.39 \text{ g}_{(\text{water})} \text{ g}_{(\text{MOF})}^{-1} \text{ day}^{-1}$, a 117% increase over the solar-driven system. This enhancement, stemming from the doubled operational time and a steady, weather-independent heat source, establishes the method as a more efficient and reliable strategy, particularly for sun-limited regions.

To elucidate the internal thermal behavior governing this thermally symbiotic system, we performed finite-element thermal simulations that trace spatiotemporal heat-transfer pathways within the coupled modules. The simulations reveal critical design parameters—including interface thermal conductivity and battery discharge profiles—that influence thermal transport efficiency and desorption performance. Together, our experimental and computational analyses demonstrate a self-regulating, closed-loop thermal integration strategy that refunctionalizes waste heat from energy storage systems to drive decentralized water harvesting. This is a report to realize AWH and BTM as a single thermally symbiotic design. By harmonizing disparate thermal requirements through shared heat infrastructure, this framework offers a versatile blueprint for integrating heat-intensive and heat-generating subsystems. The broader principle of “thermal symbiosis” introduced here can be extended to a wide range of energy and environmental technologies seeking improved thermal efficiency through intra-system heat circulation.

Results

A thermally symbiotic BTM–AWH architecture

As illustrated in Fig. 1a, AWH with MOFs typically involves a two-step cycle: adsorption and desorption. Initially, at lower ambient temperatures, the MOF adsorbs water vapour, reaching a hydrated state. Subsequently, during the regeneration phase, the temperature is elevated—typically above 40 °C—inducing desorption and returning the MOF to a dehydrated state^{39,44}. This desorption step is energy-intensive, often requiring an external heat source. Interestingly, this heat demand aligns with the thermal management characteristics of lithium-ion batteries. During operation, batteries generate significant

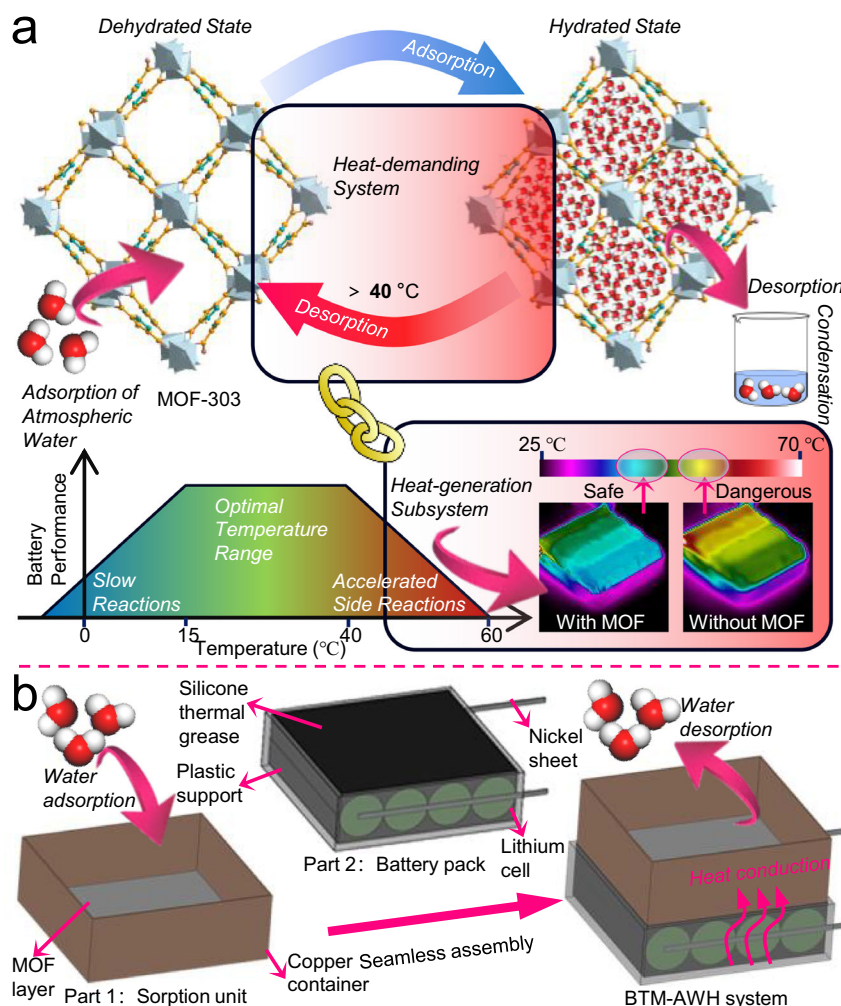


Fig. 1 | Concept and configuration of the thermally symbiotic BTM-AWH system. a Conceptual illustration of the complementary thermal roles of MOF-based atmospheric water harvesting and battery operation. The MOF-303 sorption unit adsorbs atmospheric water at low temperature and releases water upon heating above 40 °C, whereas the battery pack generates heat during discharge. Infrared

images compare battery packs with and without the MOF-based sorption unit.

b Components, assembly and working principle of the BTM-AWH system, in which a MOF-303 sorption unit is thermally coupled to a lithium-ion battery pack through silicone thermal grease to enable heat transfer for water desorption.

waste heat, which must be dissipated to maintain temperatures below a safe operating threshold of 55 °C. This creates a complementary supply-demand relationship: the AWH system requires heat for desorption, while the battery produces excess heat that must be removed.

We captured infrared images of battery packs with and without an integrated MOF-based AWH system. Under identical discharge conditions, the battery pack with the BTM-AWH system exhibited a lower temperature. This temperature reduction indicates that a portion of the battery's waste heat was transferred to drive the AWH process, thereby supporting the concept of thermal symbiosis.

The system comprised two main components, as depicted in Fig. 1b. The first component, referred to as the sorption unit, consisted of a $2.5 \times 6.5 \times 7.5$ cm copper container housing a 5 g MOF-303 layer for water absorption and desorption processes (Suppl. Fig. 1). MOF-303^{32,43}, which has superior adsorption kinetics compared to other benchmark MOFs or salts under arid conditions (Suppl. Table 5), was characterized in terms of its crystal structure, microstructure, and pore size distribution (Suppl. Figs. 2–4 and Suppl. Table 1). The MOF layer reaches adsorption equilibrium within 30–40 min across 20–60% RH (Suppl. Fig. 40a). Fitted with a pseudo-first-order kinetic model, the adsorption rate constant (k_f , Suppl. Equation 6) at 30% RH and 25 °C

was determined to be 0.14 min^{-1} (Suppl. Fig. 40b, Suppl. Table 4). This copper container with high thermal conductivity facilitated heat transfer between the two components, reducing interfacial thermal resistance. The second component was a 14 Ah battery pack, assembled with four 3.5 Ah lithium cells in parallel using nickel sheets, which was further encased in plastic support for shape stabilization and partial heat insulation (Suppl. Fig. 5). To ensure optimal contact within the system, silicone thermal grease with a high thermal conductivity of approximately $5 \text{ W m}^{-1} \text{ K}^{-1}$ was utilized to fill the air gaps between the sorption unit and battery pack (Suppl. Figs. 6 and 7). The silicone layer is relatively thick and non-uniform, as illustrated in Suppl. Fig. 43. This promoted efficient heat transfer by conduction through the silicone thermal grease, which replaces the air gaps.

In operation, the sorption unit first adsorbs atmospheric water before being coupled with the discharging battery pack to achieve thermal symbiosis, involving the sorption unit recycling heat from the battery pack for water desorption, while the battery pack simultaneously dissipates heat to the sorption unit. This reciprocal heat exchange helped maintain the battery pack within a safer temperature range while concurrently facilitating water production by the MOF layer.

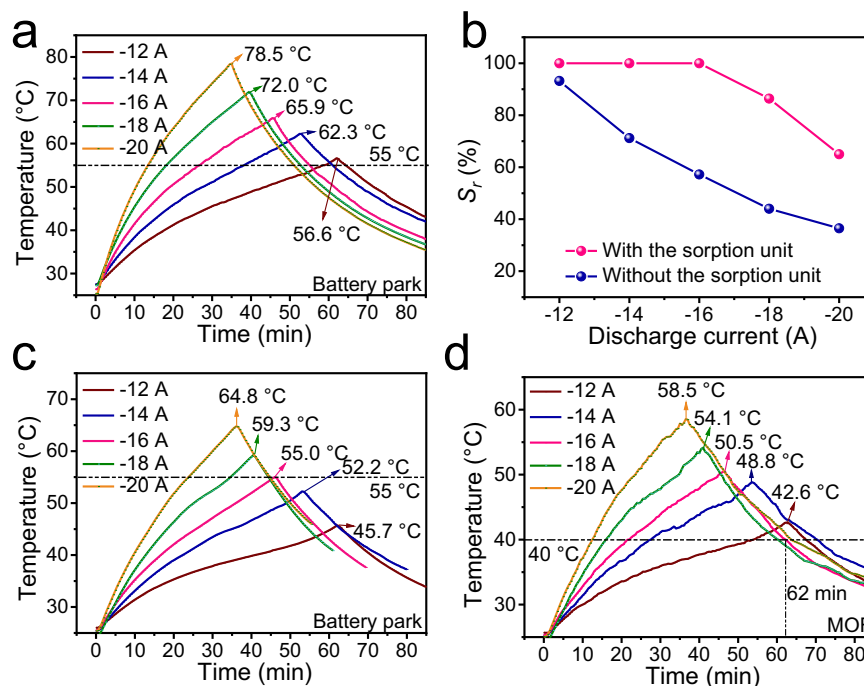


Fig. 2 | Thermal-management performance of the BTM-AWH system.

a Temperature profiles of the battery pack without the sorption unit at different discharge currents. **b** Calculated safety ratios (s_r) of the battery pack with and without the sorption unit, determined from the corresponding battery-

temperature profiles using Eq. (1); the underlying t_d , t_s , and s_r values are provided in Suppl. Table 2. **c** Temperature profiles of the battery pack with the sorption unit at different discharge currents. **d** Temperature profiles of the MOF layer coupled to the battery pack at different discharge currents.

Sorption-assisted battery thermal management

To monitor the thermal behavior, the temperature of the battery pack was recorded. When adjusting the discharge currents from -12 to -20 A, the peak temperature of the battery pack rose from 56.6 to 78.5 °C (Note: Negative currents denote the discharging process throughout this work). This increase was attributed to significant resistance heating during the discharge process, which intensified as the discharge current grew. Concurrently, the total discharge time (t_d) decreased from 62.8 to 36.9 min (Suppl. Fig. 8). To assess the battery pack's safety, the safety ratio (s_r) was introduced and calculated using Eq. 1.

$$s_r = \frac{t_s}{t_d} \quad (1)$$

t_s refers to the safe operation time of the battery pack, during which the system temperature stayed below 55 °C^{45,46}. (Suppl. Table 2) The specific measurement method of t_s , sampling frequency of temperature monitoring, and layout of measuring points are specified in Supplementary Fig. 41.

Figure 2b summarizes the s_r calculated from the corresponding battery-temperature profiles; the underlying t_d and t_s values are listed in Suppl. Table 2. s_r exhibited a decreasing trend with increasing discharge currents, registering at 93.2%, 71.2%, 57.2%, 44.0%, and 36.5% as the discharge currents varied from -12 to -20 A. In Fig. 2c, upon the assembly of the sorption unit, the peak temperature of the battery pack showed a reduction, falling within the range of 45.7–64.8 °C when subjected to discharge currents of -12 to -20 A. This marked a decrease of approximately 10 °C at each discharge current compared to configurations without a sorption unit. Consequently, s_r of the battery pack rose to 100.0%, 100.0%, 100.0%, 86.4%, and 65.0%, respectively.

Across the tested discharge currents, the incorporation of the sorption unit improved heat dissipation, lowering the battery pack's

peak temperature and reducing the risk of thermal runaway. This enhancement also increased s_r , delaying battery aging and enabling safe, stable system operation within a -12 to -16 A current range. Simultaneously, in Fig. 2d, the MOF temperature displayed a similar trend, with slightly reduced peak temperatures compared to those of the battery pack, ranging from 42.6 to 58.5 °C within a -12 to -20 A current range, indicating an effective heat transfer pathway between the battery pack and MOF. In subsequent discussions, the discharge current was set to -16 A, balancing requirements of MOF desorption and BTM – desorption occurs when the MOF temperature is higher (regeneration temperatures typically exceed 40 °C and refs. 39,44) while ensuring the battery pack remains below 55 °C. Notably, the desorption time was set to 62 min, during which the MOF temperature exceeded the regeneration threshold.

Battery waste heat drives sorbent regeneration

The sorption behavior of the sorption unit was examined as follows. MOF-303 exhibits relatively high desorption energy due to its hydrophilic functional groups, which strongly interact with water molecules^{32,43}. In Fig. 3a, MOF-303's desorption activation energy was fitted and calculated based on its desorption performance under various heating rates (Suppl. Fig. 9), using the Kissinger equation^{47,48} (Suppl. Equation 1), amounting to approximately 40 kJ mol⁻¹. This value aligned with the reported literature, indicating its high capacity to absorb heat generated by the battery pack.

In Fig. 3b, we compared the desorption process to traditional solar-driven desorption, with MOF layers pre-adsorbing water at 30% RH and 25 °C for 30 min (Suppl. Fig. 10). Under a 1.0 kW m⁻² light irradiance, the MOF layer was heated to 39 °C in roughly 20 min, due to limited light absorbance (Suppl. Fig. 11). The solar-driven MOF layer's temperature exceeded the battery pack-driven counterpart around 20 min. However, after 20 min, the battery pack-driven MOF layer displayed a higher temperature, which may lead to a superior desorption rate.

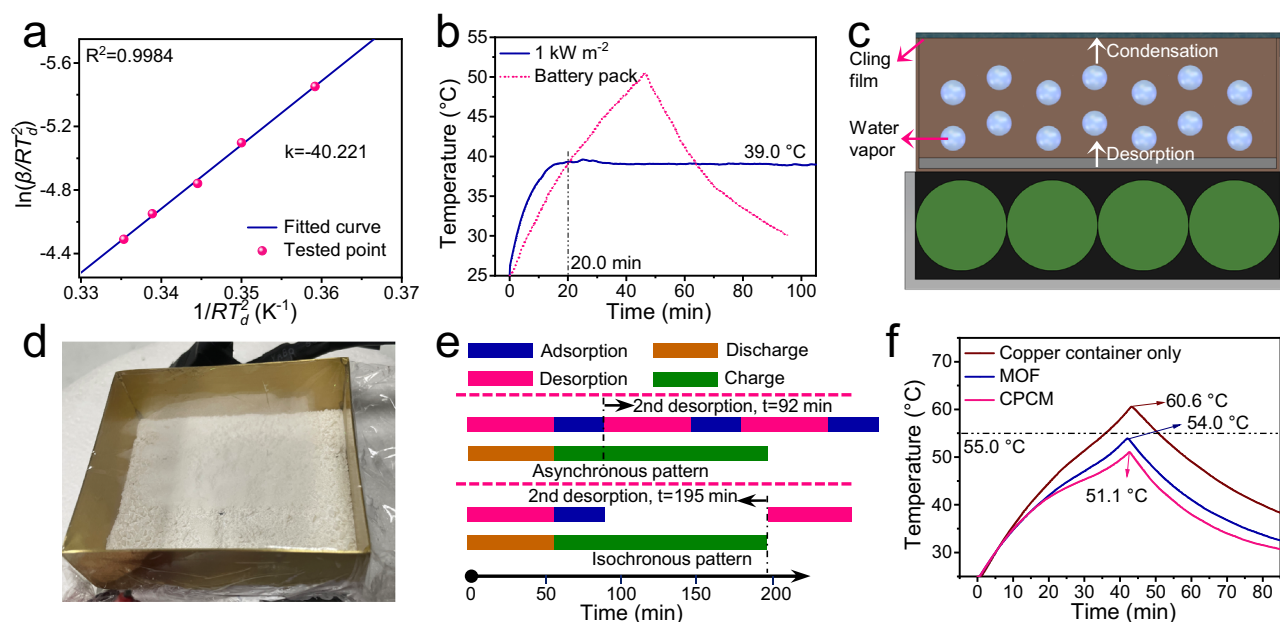


Fig. 3 | Atmospheric water harvesting enabled by battery-pack-driven sorbent regeneration. **a** Linear fit used to determine the desorption activation energy of MOF-303. **b** Temperature profiles of the MOF layer under simulated sunlight (blue solid curve, 1.0 kW m^{-2}) and battery-pack heating (pink dotted curve). The vertical dashed line at 20 min marks the approximate first crossover, after which the battery-pack-driven MOF temperature exceeds that under simulated sunlight.

c Schematic of the closed sorption unit and its water-production mechanism. **d** Digital photograph of the closed sorption unit after water generation. **e** Schematic timelines of adsorption–desorption and charge–discharge under isochronous and asynchronous operating patterns. **f** Temperature profiles of the battery pack under different thermal-management conditions.

Cling film was employed to seal the sorption element to simulate an actual AWG, creating a simple closed system as depicted in Fig. 3c. Before desorption (discharge), there was no water vapor condensation on the inner cling film (Suppl. Fig. 12). During discharge, heat was transferred from the battery pack to the MOFs for desorption, leading to a closed cavity with high water vapor partial pressure due to the abundance of desorbed water. Concurrently, the copper container's temperature increased due to its high thermal conductivity. This temperature rise increased the air temperature within the cavity due to heat transfer from the heated copper container, accompanied by convective air mixing. The temperature gradient led to cooling of the inner cling film surface, facilitating condensation of water vapor, as illustrated in Fig. 3d. This lab-scale AWG experiment demonstrated the potential for water production facilitated by the battery pack.

Battery pack-driven water production can be achieved without sunlight, broadening the application of sorbents in regions with limited sunlight and extending their daily production time to increase water yield. In this study, we initially employed an isochronous and one-to-one mapping approach, pairing one sorption unit with one battery pack during the thermal symbiosis process. To preserve battery life, the charging current was limited to 0.5 C (7 A), resulting in a charging time of approximately 133 min (Suppl. Fig. 13). As illustrated in Fig. 3e (the clearer schematic diagram is shown in Suppl. Fig. 39), one cycle consisted of a 62 min desorption and water generation period (during this period, the battery pack completed the discharge process), followed by a 30 min adsorption period at 30% RH and 25°C (Supplementary Fig. 10), allowing the sorption to equalize. Concurrently, the battery pack began recharging for 133 min. However, the battery pack's charging time exceeded the sorption unit's adsorption time, creating a timing mismatch in which the battery pack required more time before initiating the next sorption cycle. Consequently, one cycling time of the sorption unit was extended to approximately 195 min, including a 62 min water generation and a 133 min sorption unit's charge time, which reduced the daily (24 h) theoretical cycle number to only 7 times.

In contrast to the isochronous battery-to-sorbent pairing, we adopted an asynchronous configuration that eliminates downtime associated with the battery's prolonged charging period. By coordinating three battery packs (Packs 1, 2, and 3) with a single sorption unit, the system enables continuous regeneration cycles—eliminating the need to wait for the same pack to recharge. As shown in Fig. 3e, the sorption unit can complete one cycle in 92 min using a single battery pack. The three battery packs contributed to water generation during the 0–92, 92–184, and 184–276 min intervals, respectively. After completing its first discharge, Pack 1 had enough time to charge, preparing for its second water generation cycle at 276 min. This asynchronous design reduced the sorption unit's cycle time to 92 min, increasing its theoretical daily cycle count to 15, representing a 114% improvement over the isochronous configuration. The increased production frequency of the BTM-AWH system resulted in a higher daily water yield than that achievable under standard 1.0 kW m^{-2} solar irradiance, the typical condition for AWG, as conclusively demonstrated in Fig. 5.

MOF-303 is well recognized for its exceptional stability. To evaluate its cycling stability, we tested the adsorption behavior of MOF-303 after 50 sorption cycles; the detailed procedure is provided in the Supporting Information (Supplementary Fig. 14). Notably, the adsorption kinetics of MOF-303 showed minimal change over 50 cycles, as supported by nearly identical pore-size distributions, preserved crystalline frameworks, and stable granular morphology (Suppl. Figs. 2–4). This structural and performance consistency highlights the material's robustness and strongly indicates the long-term operational stability potential of the BTM-AWH system. Moreover, the introduction of the AWG module enhances the long-term performance of the battery pack, manifesting as superior electrochemical characteristics (Suppl. Fig. 45).

Herein, to better evaluate the thermal management performance of our system, we measured the battery heat dissipation under various conditions. Initially, a copper container was placed on the battery pack to enhance heat transfer and facilitate heat dissipation. In Fig. 3f, under

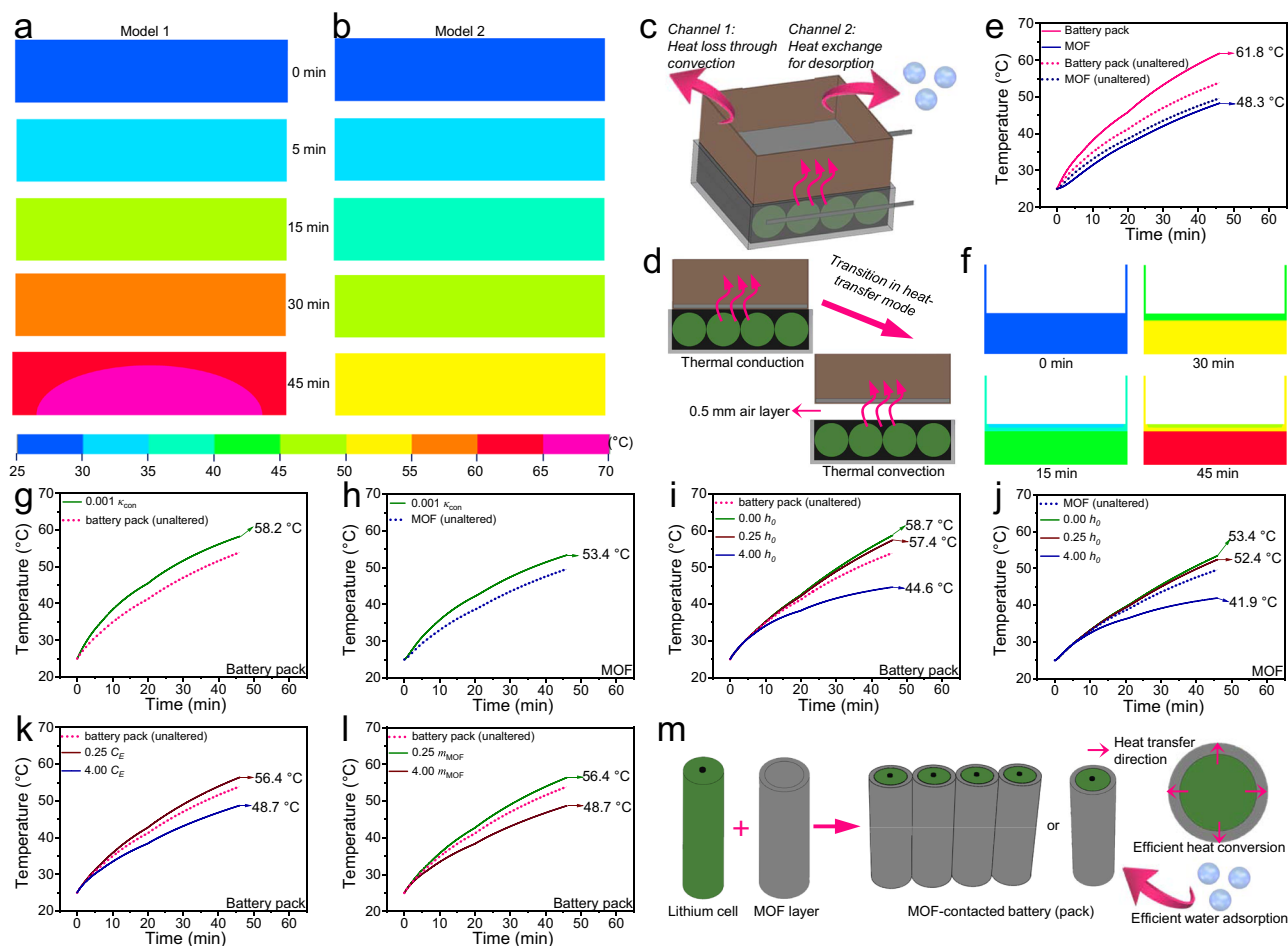


Fig. 4 | Thermo-symbiotic heat sharing in the BTM-AWH system. a, b Simulated temperature distributions of Model 1, comprising the battery pack alone (**a**), and Model 2, comprising the battery pack coupled with the MOF layer (**b**), at the indicated discharge times. **c** Schematic of the two heat-transfer channels in Model 2: heat loss to the surroundings through convection (Channel 1) and heat exchange from the battery pack to the MOF layer for desorption (Channel 2). **d** Schematic of an altered Model 2 incorporating a 0.5 mm air layer between the battery pack and MOF layer to modify the interfacial heat-transfer mode. **e** Simulated temperature profiles of the battery pack and MOF layer in the altered Model 2. **f** Corresponding simulated temperature distributions at the indicated discharge times.

a discharge current of -16 A, the pack reached a peak temperature of 60.6°C , which was slightly lower than that of the group without the copper container (65.9°C , Fig. 2a). Then, the MOF content was increased to 10 g, and this resulted in a peak temperature of 54.0°C , indicating that higher MOF loading contributes to reduced peak temperatures (compared to 55°C in the reference case, Fig. 2c). Further improvement was observed when the MOF was replaced with 10 g of expanded graphite-based eutectic CPCMs, which has a phase change temperature of 38.25°C and an enthalpy of 179.60 kJ kg^{-1} (Suppl. Fig. 15). This configuration reduced the peak temperature to 51.1°C , suggesting that CPCMs can significantly enhance the thermal safety of the system. The improved performance is likely associated with the carbon-based conductive framework of expanded graphite, which enhances heat transfer and thermal storage from the battery pack. Comparing the three systems, we conclude that while natural convection provides considerable cooling, the incorporation of heat-absorbing materials, especially those enabling direct-contact heat exchange, further enhances heat dissipation. These results underscore the rationale for adopting a thermal symbiosis approach in BTM.

g, h Simulated temperature profiles of the battery pack (**g**) and MOF layer (**h**) after reducing the thermal conductivity of the copper container to $0.001\kappa_{\text{con}}$. **i, j** Simulated temperature profiles of the battery pack (**i**) and MOF layer (**j**) at different convective heat-transfer coefficients, h_o . **k** Simulated battery-pack temperature profiles at different equivalent specific heat capacities, C_E . **l** Simulated battery-pack temperature profiles at different MOF loadings, m_{MOF} . Solid curves in (e–l) show the indicated altered-model results, whereas dotted curves denote the corresponding unaltered temperature profiles. **m** Conceptual illustration of an MOF-contacted battery configuration for thermo-symbiotic heat sharing, efficient heat conversion and water adsorption.

Thermo-symbiotic heat sharing pathways

During the 46 min heating period, both the MOF and battery pack reached peak temperatures (Fig. 2c, d). To analyze the thermo-symbiotic mechanism of this system, a thermal simulation was conducted, observing the temperature tendencies and peak temperatures of each component. The model mesh, calculation equations, and assumed conditions are detailed in Supplementary Note 16 (Suppl. Figs. 16–25). Firstly, a pure battery pack model without the sorption unit was established, termed Model 1 (Suppl. Fig. 18). In this model, the experimental and simulated battery pack temperatures demonstrated a strong correlation, indicated by a low deviation value (D_v , Suppl. Equation 4) ranging from -0.57 to 1.00 K and a low deviation rate (D_r , Supplementary Equation 5) ranging from -0.17% to 0.33% . This result confirms the model's excellent reliability (Suppl. Figs. 19–21).

On this basis, we analyzed the temperature profile of the pure battery pack cross-section (Suppl. Fig. 26) that passes through the geometric center, as depicted in Fig. 4a. Before discharge, the battery pack appeared blue, corresponding to the ambient temperature of 25°C . Within 45 min, due to a high heating rate, the color transitioned from blue-green to an orange-red hue. Interestingly, the battery pack

demonstrated a consistent temperature distribution at each recorded interval, suggesting limited heat exchange with the surrounding environment, even at the edge profile where convective heat transfer would typically be more pronounced. The only exception occurred at the 45 min mark. This limited heat dissipation, leading to an increase in the battery pack's temperature. It was found that the battery pack temperature decreased when the convective heat-transfer coefficient (h_o) was increased via simulation, moving towards a forced convection state (Suppl. Fig. 27). However, achieving such enhanced convective heat transfer would typically require additional energy input⁴⁹.

Next, with equal battery pack heating power input and consistent environmental factors, the sorption unit was integrated into the battery pack, resulting in Model 2, namely the thermo-symbiotic system (Suppl. Fig. 22). The cross-section of the Model 2 system also passed through a geometric center, as illustrated in Supplementary Fig. 26. To simplify calculations while accounting for desorption heat and sensible heat, the equivalent specific heat capacity (C_E) method was employed to set the MOF's specific heat capacity, as detailed in Suppl. Figs. 16 and 17. Similarly, the simulated and experimental temperature curves of the battery pack and MOF in Model 2 exhibited a strong correlation, indicated by low D_v values ranging from -1.08 to 0.59 K and -0.74 to 0.38 K, and low D_r values ranging from -0.33% to 0.20% and -0.23% to 0.13% , respectively (Suppl. Figs. 23–25). These results demonstrate the accuracy of the method and the reliability of the model.

In Model 2, the battery pack exhibited lower temperatures than Model 1 at nearly all recorded intervals, as indicated by the darker colors in the thermal profile of Fig. 4b. The only exception occurred at the 5-min mark. Having the same convection setting as Model 1, the battery pack in Model 2 maintained a consistent color at each recorded interval, suggesting that convection plays a limited role and that other heat dissipation mechanisms may also contribute. The temperature profiles of the integrated components (copper container and MOF layer) were observed (Suppl. Figs. 28–30), leading to the preliminary prediction of two channels for the thermo-symbiotic mechanism, as shown in Fig. 4c. First, the heat generated during battery discharge is transferred to the copper container through the silicone thermal grease layer, aided by the high thermal conductivity of copper, as evidenced by the temperature profile of the system in Suppl. Fig. 28. Next, the heat, with two channels, is dissipated through convection to the surroundings and utilized for MOF regeneration. To confirm this hypothesis, comparative experiments were simulated by adjusting the heat-transfer mode or parameters.

In Fig. 4d, Model 2 was reconstructed by inserting a 0.5 mm air layer between the sorption unit and the battery pack, replacing efficient conductive heat transfer with a low-conductivity air gap and disrupting the close-contact structure. As anticipated, in Fig. 4e, the temperature difference between the MOF layer and the battery pack was greater than in the unaltered Model 2 (dotted curves represent the components' temperature curves in the unaltered Model 2). Specifically, the peak temperature of the battery pack increased to 61.8°C , while the peak temperature of the MOF layer decreased to 48.3°C . In this scenario, both the BTM and MOF desorption effects were diminished due to the thermal resistance of air with low thermal conductivity. In Fig. 4f (the temperature profile matched the temperature-dependent color axis in Fig. 4a, b), a distinct color transition, functioning as a “temperature speed bump,” was observed in the air zone of this system, providing qualitative visual support for the reduction of the two thermal effects. This change in the altered Model 2 highlighted the importance of close contact between system components.

κ of the copper container was adjusted to 0.001 of its original value (denoted as κ_{con} , $387 \text{ W m}^{-1} \text{ K}^{-1}$), simulating the replacement of the copper container with materials having lower heat-transfer efficiency, such as a plastic-based (approximately $0.3 \text{ W m}^{-1} \text{ K}^{-1}$) container. As expected, this container exhibited different and higher-

temperature colors at each recorded time compared to the unaltered Model 2, due to slower heat dissipation (Suppl. Fig. 31). Consequently, in Fig. 4g, the BTM effect was reduced compared to the unaltered Model 2, resulting in an increased peak temperature of the battery pack at 58.2°C , indicating delayed conduction of the immense heat generated during discharge. Interestingly, as shown in Fig. 4h, the heat accumulation in the MOF also increased, with a peak temperature reaching 53.4°C . In this scenario, the overall system's heat dissipation through air convection, closely related to the container's heat-transfer capability, was diminished. In effect, the containers with $0.001 \kappa_{\text{con}}$ acted as insulation, forming a natural thermal barrier that prevented excessive heat diffusion in the MOF, contributing to MOF desorption. Simultaneously, the thin thickness (1 mm, Suppl. Fig. 1) of the container resulted in limited thermal resistance between the MOF layer and the battery pack, further promoting the MOF temperature increase. Notably, the material transformation in the container suggested a positive correlation change in the temperatures of the MOF layer and battery pack, potentially leading to two outcomes: improving desorption efficiency at the cost of reduced maximum working current limits, or enhancing the container's convection to ensure the BTM effect.

In an altered Model 2, h_o of the copper container was modified. As depicted in Fig. 4i, j, when setting $h_o = 0$, corresponding to the removal of convective heat loss at the boundary, the peak temperatures of the battery pack and MOF increased to 58.7 and 53.4°C , respectively, due to reduced heat dissipation in the system. Similarly, with a reduced convective effect of $0.25 h_o$, the peak temperatures rose to 57.4 and 52.4°C , respectively. The limitation of heat loss through Channel 1 (Fig. 4c) increased the thermal load on MOF desorption, which may reduce the BTM capacity. When h_o was quadrupled, the peak temperatures decreased to 44.6 and 41.9°C , respectively, indicating an improved BTM effect under enhanced convective heat transfer conditions. However, MOF desorption was severely hampered. The infrared thermal imaging photos detected the temperature variation of the surroundings around the assembly device (Suppl. Fig. 32). Both the experimental and simulation results demonstrated that heat dissipation through convection by the copper container is crucial for BTM, further corroborating the validity of Channel 1.

To validate Channel 2, the heat exchange for desorption, MOF parameters were altered in a reconstructed Model 2. In Fig. 4k, increasing equivalent specific heat (C_E , Suppl. Figs. 16, 17) to four times its baseline value reduced the battery pack's peak temperature to 48.7°C , while reducing it to one-quarter increased the peak temperature to 56.4°C . MOF-303 with good hydrophilicity that enabled water adsorption even in low-RH regions (Suppl. Fig. 10), exhibited a high desorption enthalpy of approximately 700 J g^{-1} (Suppl. Fig. 16), surpassing most melting enthalpies of reported CPCMs^{50–62} (Suppl. Table 3) for BTM, ranging from 100 – 200 J g^{-1} . As expected, the battery pack temperature was affected by the total MOF mass (m_{MOF}), as shown in Fig. 4l. The peak temperatures of the battery pack were 56.4 and 48.7°C when setting $0.25 m_{\text{MOF}}$ and $4 m_{\text{MOF}}$, respectively. A higher loading of MOF enhanced the BTM effect. However, it simultaneously lowered the water production capacity per unit mass of MOF, primarily due to a reduction in average MOF temperature and an increase in mass transfer resistance. This dose-performance trade-off is illustrated in Suppl. Figs. 33, 42. Notably, because the C_E and mass collectively govern the MOF's thermal energy absorption capacity, the simulated temperature values in Fig. 4k, l remain nearly identical when both parameters are scaled by the same factor throughout the simulation.

Closed-chamber water generation

To evaluate the actual water production of the BTM-AWH system, we constructed a closed-chamber water generator (Fig. 5a, b). The main body consisted of a transparent acrylic plate (Suppl. Fig. 34), inside which the BTM-AWH unit was housed. A semiconductor cooler was

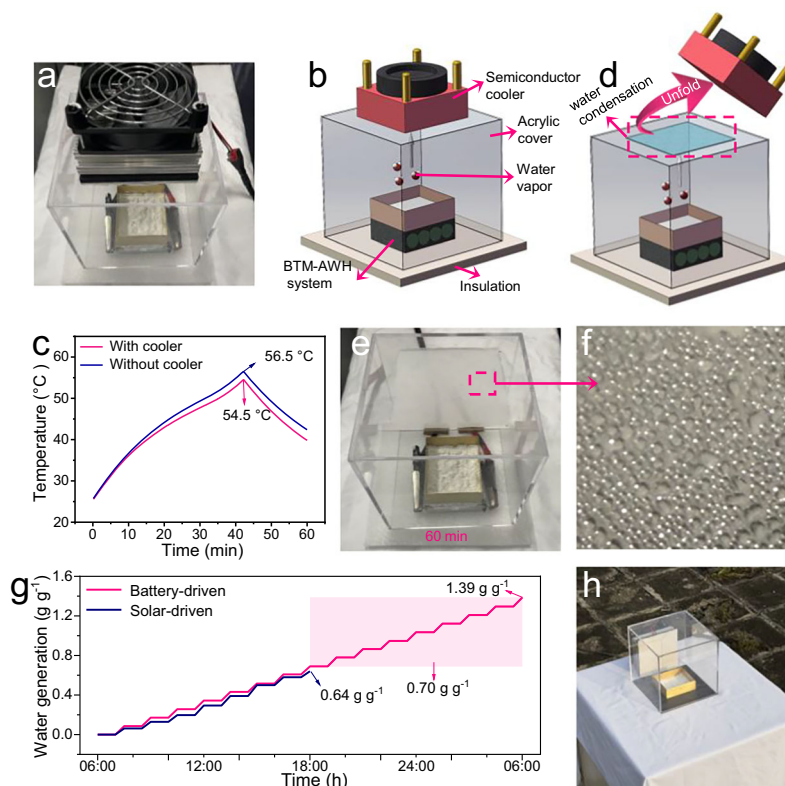


Fig. 5 | Water generation in the closed BTM-AWH system. **a** Digital photograph of the closed BTM-AWH water generator. **b** Schematic of the closed water-generation setup incorporating the BTM-AWH unit, acrylic cover, semiconductor cooler and base insulation. **c** Temperature profiles of the battery pack in the closed BTM-AWH system with and without semiconductor cooling. **d** Schematic of water-

vapour transport and condensation on the cooled inner surface of the acrylic cover. **e** Digital photograph of the closed BTM-AWH system after 60 min of operation. **f** Enlarged view of condensed water droplets on the cooled inner surface. **g** Cumulative water production of the battery-driven and solar-driven AWH systems. **h** Digital photograph of the solar-driven closed-system AWH unit.

installed on the top side of the plate to facilitate water condensation. The bottom of the chamber was insulated to minimize heat loss, maintain internal temperature, and promote a thermal gradient with the external environment. Within this system, 5.0 g of MOF was used for water harvesting. The adsorption phase was conducted for 30 min at 25 °C and 30% RH (Supplementary Fig. 10), followed by a 60 min desorption phase, during which the MOF temperature reached approximately 50 °C (Suppl. Fig. 35). Notably, as shown in Fig. 5c, the peak temperature of the battery pack was reduced to 54.5 °C despite the acrylic enclosure hindering natural convection. This temperature is lower than that observed in the open system (Fig. 2c), indicating that the active cooling from the semiconductor cooler effectively offset the heat buildup. This conclusion is supported by a control experiment in which the cooler was removed (Suppl. Fig. 36), resulting in a peak battery temperature of 56.5 °C, which exceeds the safe operating threshold. The operational process of the BTM-AWH system is shown in Fig. 5d, e. Following the simultaneous activation of the battery pack and semiconductor cooler, the cooler was temporarily displaced, revealing a distinct cooling region on the generator ceiling. Within this region, a hazy mist initially formed and subsequently coalesced into visible water droplets as the temperature gradient increased (Suppl. Fig. 37), with prominent condensation evident after 60 min (Fig. 5f). The mass of water produced was measured for each cycle. For the closed-chamber water-generation experiment, each cycle comprised a 60 min desorption stage followed by a 30 min adsorption stage, giving a 90 min cycle. Over 24 h, 16 cycles yielded a total of 6.95 g of water, corresponding to a daily production rate of $1.39 \text{ g}_{(\text{water})} \text{ g}_{(\text{MOF})}^{-1} \text{ day}^{-1}$ (Fig. 5g).

For comparison, a traditional solar-driven AWH experiment was conducted outdoors in Guangzhou on October 17, 2025, from 06:00 to

18:00. In this setup (Fig. 5h), the AWH unit was operated independently from the battery system using identical 30 min adsorption and 60-min desorption cycles. To optimize solar efficiency, the cooler was positioned on the side of the generator to minimize shading, and a black cardstock was placed at the bottom to enhance the photothermal effect, thereby raising the internal temperature to facilitate desorption. During the 12 h test period, 8 cycles were completed, collecting 3.2 g of water, which equates to a daily yield of $0.64 \text{ g}_{(\text{water})} \text{ g}_{(\text{MOF})}^{-1} \text{ day}^{-1}$ (Supplementary Fig. 38). In this experiment, the BTM-AWH system achieved a 117% higher daily water yield than the solar-driven AWH system. Also, the BTM-AWH system demonstrates a clear advantage. Even when operated for only 12 h, it achieved a water yield of $0.70 \text{ g}_{(\text{water})} \text{ g}_{(\text{MOF})}^{-1} \text{ day}^{-1}$, surpassing the yield of the solar-driven system over the same duration. This superior production rate, combined with the ability to operate continuously without being constrained by daylight, makes the BTM-AWH system significantly more effective. Under the tested conditions, the steady thermal energy supplied by the battery enabled continuous operation independent of daylight and produced a higher water yield than the solar-driven system, suggesting a potential advantage in regions with limited sunlight. Finally, we propose a hybrid approach (Fig. 5h) that utilizes solar-driven AWH for 12 h during the day and battery-driven AWH for 12 h at night. This combined method achieves a total daily water yield of $1.34 \text{ g}_{(\text{water})} \text{ g}_{(\text{MOF})}^{-1} \text{ day}^{-1}$ (comprising $0.64 \text{ g}_{(\text{water})} \text{ g}_{(\text{MOF})}^{-1} \text{ day}^{-1}$ from the daytime and $0.70 \text{ g}_{(\text{water})} \text{ g}_{(\text{MOF})}^{-1} \text{ day}^{-1}$ from the nighttime), representing a 109% increase compared to the conventional daytime-only solar-driven AWH system. This result demonstrates that thermal symbiosis can significantly enhance traditional solar water harvesting by enabling continuous 24-h operation.

Under dynamic conditions (e.g., vibration or tilting), liquid condensate may leak or accumulate near electrical contacts, thereby posing a risk of short circuits or insulation failure. To mitigate this risk in future practical designs, the following safety measures are proposed: First, electrical contacts should be insulated and physically separated from the condensate pathway to reduce the risk of moisture exposure. Second, structural optimization of the prototype is suggested: as illustrated in Suppl. Fig. 44, securing the BTM-AWH assembly with a fixed cover can help mitigate vibration and tilting while also shielding the assembly from dripping water. Furthermore, elevating the assembly on a platform would allow any condensate to flow safely beneath it, ensuring no direct contact between water and electrical components.

Discussion

This study introduces a thermo-symbiotic system in which waste heat generated during battery discharge drives MOF-based atmospheric water harvesting, while the sorption unit concurrently moderates the battery temperature. The coupled system effectively utilizes battery waste heat to drive MOF desorption, maintaining safe operating temperatures (below 55 °C) under high discharge currents while simultaneously achieving a substantial enhancement in water yield compared to solar-driven analogues. The implementation of an asynchronous operational mode further enables continuous operation without solar heating for sorbent regeneration. These results collectively demonstrate a scalable strategy for synergistic energy and water management. Beyond the specific configuration presented, the “thermo-symbiosis” framework is generalizable. Its core principle—repurposing low-grade waste heat for a useful thermal demand—can be extended to larger battery systems (e.g., in electric vehicles or stationary storage) and is compatible with other sorbents featuring low regeneration temperatures. This design offers a practical solution for off-grid or resource-constrained environments, such as telecommunication stations or remote facilities, where concurrent thermal management and water supply are critical. Looking forward, this work opens promising avenues for material innovation (e.g., composites with enhanced thermal properties) and system-level integration with renewable energy sources, paving the way toward fully autonomous, sustainable water-energy systems. Thus, the present study not only establishes a functional prototype but also provides a versatile blueprint for synergistic resource recovery in a wide range of applications.

Methods

Materials

Aluminum nitrate nonahydrate ($\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, AR) and sodium acetate trihydrate ($\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$, AR) were purchased from Tianjin Kemiou Chemical Reagent Co., Ltd. Sodium hydroxide (NaOH , AR) was purchased from Guangzhou Chemical Reagent Factory. Pyrazole-3,5-dicarboxylic acid ($\text{C}_5\text{H}_4\text{N}_2\text{O}_4 \cdot \text{H}_2\text{O}$, AR), sodium thiosulfate pentahydrate ($\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$, AR) and strontium chloride hexahydrate ($\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$, AR) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. The 3.5 Ah lithium cells were purchased from Shenzhen Jingdou Electronic Technology Co., Ltd. Thermal conductive silicone was purchased from Nuofeng Electronic Technology Co., Ltd. Nickel sheet was purchased from Jiangsu Huiyu Alloy Technology Co., Ltd.

Preparation of MOF-303

Dissolve 1.88 g $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ in 10 mL deionized water, denoted as Solution-1. Dissolve 0.78 g $\text{C}_5\text{H}_4\text{N}_2\text{O}_4 \cdot \text{H}_2\text{O}$ and 0.6 g NaOH into 15 mL deionized water, denoted as Solution-2. Mix two solutions by pouring Solution-2 drop-by-drop into Solution-1. Stir the mixture and place it at 100 °C for 24 h, and clean the residue with deionized water twice. Dry the residue in a 120 °C vacuum oven for 24 h.

Preparation of the CPCM

A PCM solution was prepared by dissolving 7.2 g $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$, 2.8 g $\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$, and 0.2 g $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$ in a sealed glass bottle. The mixture was stirred at 70 °C for 3 h. Subsequently, 8.5 g of the resulting liquid PCM was added dropwise to 1.5 g of EG under physical stirring to form the CPCM.

Program setting of the battery pack

The battery pack discharges at constant currents of -12, -14, -16, -18, and -20 A until the voltage decreases to 2.7 V. The battery pack is then charged at a constant current of 7 A until the voltage reaches 4.25 V, followed by constant-voltage charging at 4.25 V until the current decreases to 0.7 A.

Data availability

Numerical data supporting the quantitative results reported in this Article are provided in the accompanying Source Data file. Additional experimental details and supporting results are available in the Article and its Supplementary Information. Source data are provided with this paper.

References

- Blackmer-Raynolds, L. & Sampson, T. R. Overview of the gut microbiome. *Semin. Neurol.* **43**, 518–529 (2023).
- Qin, J. et al. A human gut microbial gene catalogue established by metagenomic sequencing. *Nature* **464**, 59–65 (2010).
- Ali, M. A., Al-Farga, A. & Seddik, M. A. The positive impact of honeybee activity on fennel crop production and sustainability. *Sci. Rep.* **14**, (2024).
- Parker, A. J., Williams, N. M. & Thomson, J. D. Specialist pollinators deplete pollen in the spring ephemeral wildflower *Claytonia virginica*. *Ecol. Evol.* **6**, 5169–5177 (2016).
- Sarabeev, V. et al. Symbiotic species diversity can explain invasion success and host-parasite system stability: The case of gammarid hosts. *Ecosphere* **15**, (2024).
- Kihara, K. et al. Global/temporal gene expression analysis of *Escherichia coli* in the early stages of symbiotic relationship development with the cellular slime mold *Dictyostelium discoideum*. *Biosystems* **96**, 141–164 (2009).
- Liu, Y. X. et al. 5.1 μm Ion-Regulated Rigid Quasi-Solid Electrolyte Constructed by Bridging Fast Li-Ion Transfer Channels for Lithium Metal Batteries. *Adv. Mater.* **36**, (2024).
- Xi, G. et al. Polymer-Based Solid Electrolytes: Material Selection, Design, and Application. *Adv. Funct. Mater.* **31**, (2021).
- Yang, S. J. et al. Thermally Stable Polymer-Rich Solid Electrolyte Interphase for Safe Lithium Metal Pouch Cells. *Angewandte Chemie-Int. Ed.* **61**, (2022).
- Li, Y. et al. Thermal runaway mechanism of lithium-ion battery with $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$ cathode materials. *Nano Energy* **85**, (2021).
- Wang, Y. et al. Reductive gas manipulation at early self-heating stage enables controllable battery thermal failure. *Joule* **6**, 2810–2820 (2022).
- Tran M. K. et al. A Review of Lithium-Ion Battery Thermal Runaway Modeling and Diagnosis Approaches. *Processes* **10**, (2022).
- Chen, Y. Q. et al. A review of lithium-ion battery safety concerns: The issues, strategies, and testing standards. *J. Energy Chem.* **59**, 83–99 (2021).
- Jiang, Z. Y., Li, H. B., Qu, Z. G. & Zhang, J. F. Recent progress in lithium-ion battery thermal management for a wide range of temperature and abuse conditions. *Int. J. Hydrog. Energy* **47**, 9428–9459 (2022).
- Zhang, J. Y. et al. Advanced thermal management system driven by phase change materials for power lithium-ion batteries: A review. *Renew. Sustain. Energy Rev.* **159**, (2022).

16. Lin, J. Y., Liu, X. H., Li, S., Zhang, C. & Yang, S. C. A review on recent progress, challenges and perspective of battery thermal management system. *International J. Heat Mass Transfer* **167**, (2021).
17. Tu, Y. D., Wang, R. Z., Zhang, Y. N. & Wang, J. Y. Progress and Expectation of Atmospheric Water Harvesting. *Joule* **2**, 1452–1475 (2018).
18. Wang, M. L. et al. Towards a better understanding of atmospheric water harvesting (AWH) technology. *Water Res.* **250**, (2024).
19. Chen, W. C. et al. Nighttime Atmospheric Water Harvesting Enabled by Solar Prestorage Using a Phase-Change Thermal Storage System. *Adv. Funct. Mater.* (2025).
20. Shi, W., Guan, W. X., Lei, C. X. & Yu, G. H. Sorbents for Atmospheric Water Harvesting: From Design Principles to Applications. *Angewandte Chemie-International Edition* **61**, (2022).
21. Xu, W. T. & Yaghi, O. M. Metal-Organic Frameworks for Water Harvesting from Air, Anywhere, Anytime. *Acs Cent. Sci.* **6**, 1348–1354 (2020).
22. Nguyen, H. L. et al. A Porous Covalent Organic Framework with Voided Square Grid Topology for Atmospheric Water Harvesting. *J. Am. Chem. Soc.* **142**, 2218–2221 (2020).
23. Chen, B. et al. All-natural, eco-friendly composite foam for highly efficient atmospheric water harvesting. *Nano Energy* **110**, (2023).
24. Ahrestani, Z., Sadeghzadeh, S. & Emrooz, H. B. M. An overview of atmospheric water harvesting methods, the inevitable path of the future in water supply. *Rsc Adv.* **13**, 10273–10307 (2023).
25. Xu, J. X. et al. Ultrahigh solar-driven atmospheric water production enabled by scalable rapid-cycling water harvester with vertically aligned nanocomposite sorbent. *Energy Environ. Sci.* **14**, 5979–5994 (2021).
26. Lei, C. X. et al. Polyzwitterionic Hydrogels for Efficient Atmospheric Water Harvesting. *Angewandte Chemie-Int. Ed.* **61**, (2022).
27. Zhao, F. et al. Super Moisture-Absorbent Gels for All-Weather Atmospheric Water Harvesting. *Adv. Mater.* **31**, (2019).
28. Xu, J. X. et al. Efficient solar-driven water harvesting from arid air with metal-organic frameworks modified by hygroscopic salt. *Angew. Chem. -Int. Ed.* **59**, 5202–5210 (2020).
29. Li, R. Y. et al. Hybrid hydrogel with high water vapor harvesting capacity for deployable solar-driven atmospheric water generator. *Environ. Sci. Technol.* **52**, 11367–11377 (2018).
30. Vartanian, A. Designing desorption. *Nat. Rev. Mater.* **9**, 680–680 (2024).
31. Chen, W. C. et al. A functionally asymmetric janus hygro-photothermal hybrid for atmospheric water harvesting in arid regions. *Small* **20**, (2024).
32. Hanikel, N. et al. Evolution of water structures in metal-organic frameworks for improved atmospheric water harvesting. *Science* **374**, 454–45 (2021).
33. Kim, H. et al. Adsorption-based atmospheric water harvesting device for arid climates. *Nat. Commun.* **9**, (2018).
34. Luo, F. et al. High-efficient and scalable solar-driven MOF-based water collection unit: From module design to concrete implementation. *Chem. Eng. J.* **465**, (2023).
35. Wu, Q. N., Su, W., Li, Q. Q., Tao, Y. L. & Li, H. Q. Enabling Continuous and Improved Solar-Driven Atmospheric Water Harvesting with Ti3C2-Incorporated Metal-Organic Framework Monoliths. *Acs Appl. Mater. Interfaces* **13**, 38906–38915 (2021).
36. Hanikel, N., Prévot, M. S. & Yaghi, O. M. MOF water harvesters. *Nat. Nanotechnol.* **15**, 348–355 (2020).
37. Chen, W. C. et al. Enhancing Atmospheric Water Harvesting of MIL-101 (Cr) MOF Sorbent with Rapid Desorption Enabled by Ni-Ni3S2 Photothermal Bridge. *Adv. Funct. Mater.* **34**, (2024).
38. Xu, J. X. et al. All-in-one hybrid atmospheric water harvesting for all-day water production by natural sunlight and radiative cooling. *Energy Environ. Sci.* **17**, (2024).
39. Qi, H. S. et al. An Interfacial Solar-Driven Atmospheric Water Generator Based on a Liquid Sorbent with Simultaneous Adsorption-Desorption. *Advanced Materials* **31**, (2019).
40. Wang, H. M. et al. Microtubular PEDOT-Coated Bricks for Atmospheric Water Harvesting. *Acs Appl. Mater. Interfaces* **13**, 34671–34678 (2021).
41. Tao, Y. L., Sun, J. F., Wu, Q. N., Zhu, D. R. & Li, H. Q. Tubular metal-organic framework membranes enabling exceptional atmospheric water harvesting by fast air permeation. *Chem. Eng. J.* **461**, (2023).
42. Wang, Q., Jiang, B., Li, B. & Yan, Y. Y. A critical review of thermal management models and solutions of lithium-ion batteries for the development of pure electric vehicles. *Renew. Sustain. Energy Rev.* **64**, 106–128 (2016).
43. Hanikel, N. et al. MOF Linker Extension Strategy for Enhanced Atmospheric Water Harvesting. *Acs Cent. Sci.* **9**, 551–557 (2023).
44. Zheng, Z., Hanikel, N., Lyu, H. & Yaghi, O. M. Broadly Tunable Atmospheric Water Harvesting in Multivariate Metal–Organic Frameworks. *J. Am. Chem. Soc.* **144**, 22669–22675 (2022).
45. Luo, M. Y., Lin, X. M., Ling, Z. Y., Zhang, Z. G. & Fang, X. M. An electric conductive wide-temperature flexible phase change material for all-climate battery thermal management. *Appl. Thermal Eng.* **256**, (2024).
46. Huang, Y. F. et al. A Novel Room-Temperature Flexible Phase Change Material for Solar Energy Photothermal Conversion and Battery Thermal Management. *Acs Sustain. Chem. Eng.* **12**, 4662–4675 (2024).
47. Khutia, A., Rammelberg, H. U., Schmidt, T. J., Henninger, S. K. & Janiak, C. Water Sorption Cycle Measurements on Functionalized MIL-101Cr for Heat Transformation Application. *Chem. Mater.* **25**, 790–798 (2013).
48. Luo, F. et al. Symbiotic defect-reinforced bimetallic MOF-derived fiber components for solar-assisted atmospheric water collection. *Water Res.* **259**, 121872 (2024).
49. Liu, C. C. et al. Phase Change Materials Application in Battery Thermal Management System: A Review. *Materials* **13**, (2020).
50. Wang, Z. Z., He, Y. R., Cheng, G. & Tang, T. Q. Thermal characteristics of a flame-retardant composite phase change material for battery thermal management. *Appl. Thermal Eng.* **243**, (2024).
51. Li, Y. J. et al. Flexible phase change composites with multiple colors and reversible thermochromic for temperature indication and battery thermal management. *Composites Sci. Technol.* **249**, (2024).
52. Li, J. Y. et al. Superhydrophobic multi-shell hollow microsphere confined phase change materials for solar photothermal conversion and energy storage. *Appl. Energy* **365**, (2024).
53. Verma, A., Shashidhara, S. & Rakshit, D. A comparative study on battery thermal management using phase change material (PCM). *Therm. Sci. Eng. Prog.* **11**, 74–83 (2019).
54. Vélez, C., Khayet, M. & Ortiz de Zárate, J. M. Temperature-dependent thermal properties of solid/liquid phase change even-numbered n-alkanes: n-Hexadecane, n-octadecane and n-eicosane. *Appl. Energy* **143**, 383–394 (2015).
55. Jiang, G. W., Huang, J. H., Fu, Y. S., Cao, M. & Liu, M. C. Thermal optimization of composite phase change material/expanded graphite for Li-ion battery thermal management. *Appl. Therm. Eng.* **108**, 1119–1125 (2016).
56. Deng, J. et al. Flexible composite phase-change material with shape recovery and antileakage properties for battery thermal management. *Acs Appl. Energy Mater.* **4**, 13890–13902 (2021).
57. Wu, W. X. et al. Role of natural convection and battery arrangement for phase change material based battery thermal management unit. *J. Energy Storage* **52**, (2022).
58. Jia, X. Z., Zhang, B., Chen, C., Fu, X. & Huang, Q. Immobilization of chitosan grafted carboxylic Zr-MOF to porous starch for sulfanilamide adsorption. *Carbohydrate Polym.* **253**, (2021).

59. Parkerson, Z. J., Le, T., Das, P., Mahmoodi, S. N. & Esfahani, M. R. Cu-MOF-Polydopamine-Incorporated Functionalized Nanofiltration Membranes for Water Treatment: Effect of Surface Adhesive Modification Techniques. *Acs EsT. Water* **1**, 430–439 (2021).
60. Yu, Z. P., Zhang, J. K. & Pan, W. G. A review of battery thermal management systems about heat pipe and phase change materials. *J. Energy Storage* **62**, (2023).
61. Pan, M. & Zhong, Y. Experimental and numerical investigation of a thermal management system for a Li-ion battery pack using cutting copper fiber sintered skeleton/paraffin composite phase change materials. *Int. J. Heat. Mass Transf.* **126**, 531–543 (2018).
62. Sanker, S. B. & Baby, R. Phase change material based thermal management of lithium ion batteries: A review on thermal performance of various thermal conductivity enhancers. *J. Energy Storage* **50**, (2022).

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Author contributions

W. Chen, M. Luo, and Y. Tan conceived the initial research idea. W. Chen, L. Yan, and T. Liao fabricated the devices. W. Chen, W. Liu, F. Liu, Z. Chen, and J. Yao synthesized the MOF and CPCM materials. W. Chen and X. Liang performed the characterization. S. K. Ravi, W. Chen, Y. Ke, A. Suwardi, and X. Liang contributed to data analysis, interpretation, and scientific discussion. W. Chen and M. Luo performed the heat-transfer simulations. Y. Fang, S. Wang, A. C. Y. Yuen, S. C. Tan, and S. K. Ravi guided the research. S. K. Ravi supervised the project.

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Competing interests

The authors declare no competing interests.

Additional information

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